

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol

2026-07-17

Source: `results/imo-2026/gpt-5.6-sol/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Approaches tried

- Normalize all angles by θ ; use a branch-stable nonintegrality invariant for necessity, and for integral total use an integer-crossing cut followed by descent of a positive integral angle label — worked.

Current best

The complete characterization is

$$\boxed{\theta = 180^\circ/n \quad \text{for an integer } n \geq 2.}$$

The proof derives the exact two-child transition, gives Shan-Yu a safe-child counterstrategy when $180^\circ/\theta$ is nonintegral, and gives Mulan a finite forcing strategy when it is integral.

Full proof

Set

$$s = \frac{180^\circ}{\theta}.$$

Using the **change-of-variables/reformulation technique** in the knowledge base, represent a triangle whose angles are $a\theta, b\theta, c\theta$ by (a, b, c) . The angle-sum theorem gives

$$a + b + c = s. \quad (1)$$

We first derive the transition caused by a cut. Relabel the current triangle as ABC so that Mulan's chosen perimeter point P lies in the open side BC , and the cut joins P to the opposite vertex A . Write

$$\angle A = a\theta, \quad \angle B = b\theta, \quad \angle C = c\theta, \quad \angle BAP = x\theta.$$

Because AP is strictly inside $\angle A$, $0 < x < a$ and $\angle PAC = (a - x)\theta$. By **angle chasing** from the knowledge base,

$$\angle APB = 180^\circ - (x + b)\theta = (a + c - x)\theta,$$

and

$$\angle APC = 180^\circ - (a - x + c)\theta = (b + x)\theta.$$

Thus the two children have normalized triples

$$(x, b, a + c - x), \quad (a - x, c, b + x). \quad (2)$$

Conversely, every x with $0 < x < a$ defines a legal move: the ray from A making angle $x\theta$ with AB lies strictly inside $\angle A$ and, by convexity of the triangle, meets the open opposite side BC .

Necessity

Suppose $s \notin \mathbb{Z}$. Shan-Yu initially chooses an equiangular triangle, represented by $(s/3, s/3, s/3)$. All three coordinates are nonintegral, since $s/3 \in \mathbb{Z}$ would imply $s \in \mathbb{Z}$.

We use the **invariant** and **casework/exhaustion** techniques from the knowledge base. Assume a retained triangle (a, b, c) has all three coordinates nonintegral. Since their sum is the fixed nonintegral number s , all of a, b, c, s are nonintegral. In (2), the first child already has the nonintegral coordinate b , so if that child has an integral coordinate, then either x or $a + c - x$ is integral. The second child already has the nonintegral coordinate c , so if it has an integral coordinate, then either $a - x$ or $b + x$ is integral. If both children had an integral coordinate, one of the following four pairs would consist of integers:

1. $x, a - x$, whose sum is a ;
2. $x, b + x$, whose difference is b ;
3. $a + c - x, a - x$, whose difference is c ;
4. $a + c - x, b + x$, whose sum is $a + b + c = s$.

Each case contradicts the corresponding nonintegrality. Hence at least one child has all three normalized angles nonintegral. Shan-Yu retains such a child after every cut. Induction on the number of moves preserves this invariant, so no retained triangle ever has normalized angle 1. Therefore Mulan cannot guarantee victory when s is nonintegral. Necessarily,

$$\frac{180^\circ}{\theta} \in \mathbb{Z}. \quad (3)$$

Sufficiency

Now let $s = n \in \mathbb{Z}$. Since $0^\circ < \theta < 180^\circ$, $n > 1$, and hence $n \geq 2$.

First we prove an integer-crossing lemma using the knowledge base's **pigeonhole/extremal principle**. If positive nonintegral a, b, c satisfy $a + b + c = n$, then they can be ordered so that some integer r satisfies

$$b < r < a + b. \quad (4)$$

Indeed, if one coordinate exceeds 1, call it a and call either other coordinate b . Since b is nonintegral, $r = \lfloor b \rfloor + 1$ gives

$$b < r < b + 1 < a + b.$$

Otherwise all three coordinates are less than 1. Their positive integral sum is less than 3 and at least 2, so it equals 2. Choose any two as a, b and leave the third as c . Then $b < 1 < 2 - c = a + b$, so $r = 1$ proves (4).

If the current triangle has no integral normalized angle, order its coordinates according to this lemma and set $x = r - b$. By (4), $0 < x < a$, so this is a legal cut. Formula (2) shows that the second child has normalized angle $b + x = r$, while the first has normalized angle

$$a + c - x = a + b + c - r = n - r.$$

Both labels are positive: $r > b > 0$, and $r < a + b = n - c < n$. Thus whichever child Shan-Yu keeps has a positive integral normalized angle.

It remains to prove that any positive integral normalized angle forces a finite win. We use **strong induction**, equivalently the knowledge base's well-founded integer **monovariant**. Suppose the current triangle has an angle $k\theta$ with $k \in \mathbb{Z}_{>0}$. If $k = 1$, Mulan has already won. If $k > 1$, put $j = \lfloor k/2 \rfloor$. From the vertex carrying $k\theta$, draw the interior ray making angle $j\theta$ with one incident side. This is legal because $1 \leq j < k$. The two child triangles retain at that vertex the respective angles

$$j\theta \quad \text{and} \quad (k - j)\theta.$$

Both new labels lie in $\{1, \dots, k - 1\}$. Whichever child Shan-Yu keeps therefore has a smaller positive integral label. Strong induction applies, and the strict decrease also proves termination after finitely many moves.

Mulan's strategy is now exhaustive. If the current triangle has normalized angle 1, she has won. If it has a positive integral normalized angle greater than 1, she applies the descent. If it has no integral normalized angle, she makes the crossing cut, after which either retained child has a positive integral label, and then applies the descent. Thus every integer $n \geq 2$ is sufficient.

Combining necessity and sufficiency,

$$\boxed{\theta = \frac{180^\circ}{n} \quad (n = 2, 3, 4, \dots)}$$

For every displayed value, $0^\circ < 180^\circ/n < 180^\circ$ and substitution gives $180^\circ/\theta = n$, the integral-total case proved sufficient. Every other allowed θ gives nonintegral s and is defeated by Shan-Yu's invariant strategy. $\square$